

Chapter 10

The Nila Mathematics

The Nila Renaissance: Discovery of Calculus in Medieval India
(14th – 16th Century)

Contribution of Madhava: Infinite Series for π (Pi)

The Madhava-Leibniz Series

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots$$

This series calculates π with any desired precision, a monumental leap in analytical mathematics.

Sine Series & Cosine Series

Centuries before Isaac Newton and James Gregory, Madhava derived the infinite power series for trigonometric functions. He formulated what are now globally recognized as the Maclaurin/Newton series

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots, \quad \cos \theta = \frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \dots \quad (1)$$

The Yuktibhasha: Authored by Jyeshthadeva (c. 1530)

- Written by Jyeshthadeva around 1530 CE, the Yuktibhasha (meaning "Rationale in the Vernacular" or "Discourse on Rationale") is arguably the most important text to emerge from the Kerala School. It is fundamentally considered the world's first textbook on calculus.
- **The Shift to Malayalam:** For centuries, Indian mathematics was recorded in highly condensed, esoteric Sanskrit poetry. These verses were easy to memorize but incredibly difficult to decode without a master teacher. Jyeshthadeva shattered this tradition by writing the Yuktibhasha in Malayalam prose. By using the regional language, he democratized the knowledge, making advanced mathematics and astronomy accessible to a much broader audience.
- **The Focus on Yukti (Proof):** The traditional Indian mathematical style was to present a rule or a formula without explaining how it was derived. The Yuktibhasha changed the paradigm. Jyeshthadeva dedicated the entire text to providing the Upapatti (logical proofs and step-by-step derivations) for the theorems discovered by Madhava and Nilakantha over the previous century.

The Structure of the Text

A Dual Masterpiece

The *Yuktibhasha* is divided into two distinct volumes, consolidating the genius of the Kerala School.



Volume I: Mathematics (Ganita)

Covers arithmetic, algebra, and the revolutionary foundations of calculus.



Volume II: Astronomy

Applies mathematical proofs to planetary dynamics and spherical coordinates.



Foundational Arithmetic: Chapters 1–3



Chapter 1: Parikarma

Focuses on 'Logistics' or the eight fundamental mathematical operations, including addition, subtraction, multiplication, and division.



Chapter 2: Dasaprasna

The 'Ten Problems' chapter. It explores complex word problems and logistics applied to geometry and practical arithmetic.



Chapter 3: Bhinnaganita

Deep dive into the arithmetic of fractions. Covers operations, reduction to common denominators, and geometric interpretations.

Algebra and Ratios: Chapters 4–6



Chapter 4: Trairasika

The 'Rule of Three'. Essential for proportionality and solving for unknowns in ratios.



Chapter 5: Kuttakara

The study of astronomical time-keeping problems, illustrating with numbers (in katapayadi), solution of diophantine equations, etc.



Chapter 6: Paridhi-Vyasa

he core of the text's calculus achievement. It provides the rationale for the infinite series of π .

Chapter 7: Jyanayana (Sine Derivation)

Infinite Trigonometric Series

This chapter provides the proofs for the power series of Sine and Cosine functions, anticipating Taylor Series by centuries.

- Sum and difference formulas for trigonometric functions.
- Second-order approximations.
- Proofs of Brahmagupta theorems.
- Geometry of cyclic quadrilaterals.

Taylor Series Anticipated

By dividing an arc into smaller and smaller segments and applying the '**Rule of Three**' iteratively, Jyeshthadeva demonstrates the fundamental principles of **differentiation and integration**.

Geometry & Trigonometry: Addition Theorems

- **Proof of the following identities**

$$\sin(\theta \pm \phi) = \sin \theta \cos \phi \pm \cos \theta \sin \phi,$$

$$\cos(\theta \pm \phi) = \cos \theta \cos \phi \mp \sin \theta \sin \phi$$

- From Yuktibhasha, it is said that The product of two arbitrary chords is the difference of the squares of the chords associated with the half the sum and half the difference of the corresponding arcs.

$$\sin \theta_1 \sin \theta_2 = \sin^2 \left(\frac{\theta_1 + \theta_2}{2} \right) - \sin^2 \left(\frac{\theta_1 - \theta_2}{2} \right)$$

- Equivalently, one can also observe the following:

$$\sin^2 \theta_1 - \sin^2 \theta_2 = \sin(\theta_1 + \theta_2) \sin(\theta_1 - \theta_2)$$

The Sine Table: Interpolation

- **First order interpolation approximation**

$$\sin(m\epsilon + \delta)_1 = \sin m\epsilon + \delta \cos m\epsilon,$$

$$\cos(m\epsilon + \delta)_1 = \cos m\epsilon - \delta \sin m\epsilon$$

$$\sin(m\epsilon + \delta)_2 = \sin m\epsilon + \delta \left(\cos m\epsilon - \frac{\delta}{2} \sin m\epsilon \right),$$

$$\cos(m\epsilon + \delta)_2 = \cos m\epsilon - \delta \left(\sin m\epsilon + \frac{\delta}{2} \cos m\epsilon \right)$$

- **Second order interpolation approximation**

We can derive the following power series representation if we replace $\sin(\delta/2^k) = \delta/2^k, \quad \forall k,$

as

$$\sin(\theta + \delta)_{inter} = \sin \theta + \delta \cos \theta - \frac{\delta^2}{2} \sin \theta - \frac{\delta^3}{8} \cos \theta + \dots,$$
$$\cos(\theta + \delta)_{inter} = \cos \theta - \delta \sin \theta - \frac{\delta^2}{2} \cos \theta + \frac{\delta^3}{8} \sin \theta + \dots,$$



Questions?

Thank you for your attention.